

AFRAZ MOHAMMAD

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SUMMARY

Data Engineer with **5 years** of experience designing scalable **Lakehouse architectures**, **cloud-native ELT pipelines**, and **distributed data platforms** across **healthcare**, **marketing analytics**, and **financial compliance** domains. Strong foundation in **Microsoft Fabric**, **Azure Synapse**, **Snowflake**, **Apache Spark**, and **advanced SQL**, with hands-on experience building **secure**, **governed**, and **AI-ready data systems**. Collaborative partner across **8+ cross-functional stakeholders** spanning engineering, analytics, audit, and business teams to deliver **production-grade data solutions**.

PROFESSIONAL EXPERIENCE

Data Engineer | BCBS | USA

May 2025 – Present

- Architected a **Databricks (Delta Lake)** Medallion Lakehouse on **ADLS Gen2**, integrating 18 healthcare systems into governed Gold-layer models for **6M+ members**, delivered through Microsoft Fabric (DirectLake).
- Engineered incremental and streaming **ELT pipelines** with **Azure Data Factory** orchestration and Databricks (Delta Live Tables, CDC) transformations, reducing processing time from 7 to 3.8 hours across 120+ production workflows.
- Optimized Spark workloads on **Databricks (Photon Engine)**, processing 30M+ monthly healthcare transactions and reducing query latency from **9.6s to 2.4s** via partition pruning, Z-ordering, and columnar optimization.
- Implemented enterprise data governance using Microsoft Purview and Unity Catalog, enforcing **RBAC** and **Row-Level Security (RLS)** across **250+ datasets** and clearing 14 HIPAA compliance audit checkpoints.
- Built **ML-ready** feature pipelines in **PySpark** and **MLflow**, containerized with **Docker** and deployed on **AKS (Kubernetes)**, reducing model release cycles from 28 to 17 days for population health analytics.

Data Engineer / Data Analyst | Epsilon | USA

Jul 2024 – May 2025

- Migrated **5 campaign analytics ETL workflows** to Snowflake and dbt, consolidating **2M+ customer interaction records** into a unified **dimensional model** supporting marketing performance reporting.
- Built event-driven ingestion pipelines using **Kafka and AWS Glue**, processing **2.6M daily campaign events** and enabling downstream personalization and attribution systems.
- Tuned advanced **SQL (CTEs, Window Functions)** across **Amazon Redshift** during phased warehouse transition, reducing dashboard latency from **18 seconds to 6.2 seconds** for executive revenue reporting.
- Delivered **Power BI semantic models** with **DirectLake connectivity**, publishing **12 production dashboards** used by **110+ analytics** and revenue stakeholders for weekly campaign reviews.
- Established **CI/CD data pipelines** in **Azure DevOps and GitHub**, standardizing release management for **75+ data assets** and lowering deployment defects from **11 to 3 per quarter**.

Data Analyst | SAS Info Inc | USA

Jan 2024 – May 2024

- Analyzed **2M+ financial transactions** using **SQL Server (CTEs, Window Functions)** and **Python (Pandas)**, identifying **1,400 high-risk anomalies** and reducing **false-positive fraud flags by 320 cases**.
- Developed department-level **star schema data marts** in **PostgreSQL**, modeling fact and dimension tables for **4 operational teams**, enabling standardized KPI reporting within **4 business days each month**.
- Automated recurring validation and reconciliation tasks using PowerShell scripts orchestrated via Apache Airflow, eliminating **4 manual review checkpoints** and saving **16 analyst hours monthly**.
- Implemented structured **data quality controls and audit logging frameworks**, reconciling 48 reporting datasets and resolving 90+ consistency gaps prior to executive financial submissions.

Data Analyst | KPMG | IND

Jun 2019 - Jul 2022

- Conducted structured data profiling and compliance assessments using **SQL (joins, aggregations, window functions)** and Alteryx workflows across **7 advisory engagements**, analyzing **12M+ financial records** to support SOX and internal audit reviews.
- Transformed disparate SAP and Oracle ERP extracts into dimensional warehouse models (fact/dimension tables) using **SQL Server and SSIS**, enabling standardized analytics across **11 client reporting environments**.
- Developed interactive **Tableau dashboards** visualizing audit trails, exception trends, and risk indicators, supporting quarterly governance reviews led by 8 senior managers.
- Engineered **Python-based reconciliation scripts (Pandas, NumPy)** to validate large-scale ledgers, identifying discrepancies totaling **\$2.1M in misclassified transactions** across **4 transformation programs**.

- Standardized **metadata and data lineage documentation** using structured mapping templates and dependency matrices, reducing analyst onboarding cycles to **12 business days**.
- Partnered with audit, finance, and IT stakeholders to remediate **19 internal control gaps**, aligning reporting processes with **SOX and IFRS compliance standards**.
- Produced executive-ready **ad-hoc analytics in R and advanced SQL**, delivering data-backed insights for **6 board-level financial review sessions** within regulated reporting timelines.

TECHNICAL SKILLS

Programming & Querying: SQL (CTEs, Window Functions, Query Optimization), KQL (Kusto Query Language), Python (Pandas, NumPy, PySpark), Spark SQL, R, PowerShell, Bash

Data Engineering & Architecture: ETL/ELT, Lakehouse Architecture, Medallion Architecture (Bronze/Silver/Gold), Dimensional Modeling, Star/Snowflake Schema, dbt, Change Data Capture (CDC), Incremental Processing, Materialized Views, Data Contracts, Schema Evolution, Metadata Management, Data Quality Frameworks

Big Data & Streaming: Apache Spark, Delta Lake, Kafka, Kinesis, Azure Data Explorer (ADX), Distributed Data Processing, Event-Driven Architecture

Cloud & Data Platforms: Microsoft Fabric, OneLake, Azure Data Factory (ADF), Azure Synapse Analytics, Azure Data Lake Storage (ADLS Gen2), Snowflake, AWS (S3, Glue, Athena, Redshift), BigQuery, Multi-Cloud Architecture

Business Intelligence & Analytics Engineering: Power BI (DAX, DirectLake, Composite Models), Tableau, Semantic Modeling, KPI Engineering, Executive Reporting, Self-Service Analytics

Databases: SQL Server, PostgreSQL, MySQL, Columnar Storage Optimization, Data Warehousing

Orchestration & DevOps: Airflow, Azure DevOps, CI/CD Pipelines, Git, GitHub, Docker, Jira, Agile (Scrum), DevSecOps

Security & Governance: Microsoft Purview, RBAC, Row-Level Security (RLS), Data Lineage, Audit Logging, Compliance Monitoring, Microsoft Defender, Entra ID

Observability & Optimization: SLA/SLO Tracking, Log Analytics, Root Cause Analysis (RCA), Capacity Planning, FinOps Cost Optimization

AI & Advanced Analytics Enablement: Machine Learning (ML) Data Preparation, Feature Engineering Pipelines, Model-Ready Datasets, Predictive KPI Modeling

EDUCATION

MINNESOTA STATE UNIVERSITY USA	May 2024
Master of Science in Data Science	